Activity-9

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Date

Map_storytelling

Task-1 choropleth map of illiteracy (%)

```
library (ggplot2)
library (maps)
ggplot (mapdat, aes(long, lat, group = group, fill =
 + geom_polygon (colour = "white") + coord_map
("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_c (option = Inferno) +
   labs = (title = "Illiteracy rate across the"),
   fill = "Illiteracy (%)") +
    theme_minimal().
```

Task-2 Regional_analysis

```
tapply (v, d$zone, mean)

means <- tapply (v, d$zone, mean)
means.
names (which.max (means))
names(which.max (means))
(means) - min (means)
```

Task3 compare - six classification methods

```
# equal interval
  cut(v,6)
 # quantile
cut(v, quantile (v, seq(0,1), length.out =7)),
  include.lowest=TRUE).
```

Task 4:- counts versus rates:-

```
d$count <- d$pop * 1000 * d$illit /100

head[d(order(-d$count), c("state", "count")],3)

head[d(order(-d$illit), c("state", "illit")), 3).
```

Therefore maps of rates are more suitable for comparing of illiteracy across states.

Task5:-

```
south <- d[d$zone == "South",]
 mean(south$illit)
 mean(d$illit)
 mean(south$illit)/mean(d$illit)

 south(which.max(south$illit), c("state", "illit")]
 sum(south$illit > median(d$illit)).
```

Headline:- Illiteracy Rates reveal Regional Differen